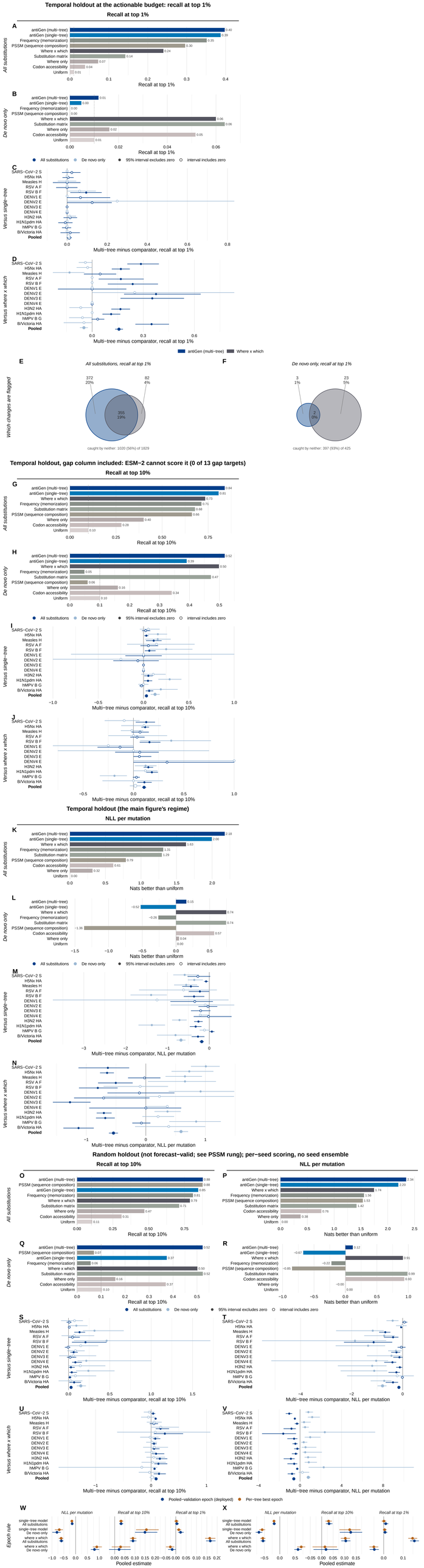
Temporal holdout at the actionable budget: recall at top 1%
Recall at top 1%
A
All substitutions
antiGen (multi−tree) 0.40
antiGen (single−tree) 0.39
Frequency (memorization) 0.35
PSSM (sequence composition) 0.30
Where x which 0.24
Substitution matrix 0.14
Where only 0.07
Codon accessibility 0.04
Uniform 0.01
0.0 0.1 0.2 0.3 0.4
Recall at top 1%
B
De novo only
antiGen (multi−tree) 0.01
antiGen (single−tree) 0.00
Frequency (memorization) 0.00
PSSM (sequence composition) 0.00
Where x which 0.06
Substitution matrix 0.06
Where only 0.02
Codon accessibility 0.05
Uniform 0.01
0.00 0.02 0.04 0.06
Recall at top 1%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
0.0 0.2 0.4 0.6 0.8
Multi−tree minus comparator, recall at top 1%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
0.0 0.3 0.6
Multi−tree minus comparator, recall at top 1%
antiGen (multi−tree)
Where x which
E
All substitutions, recall at top 1%
372
20%
82
4%
355
19%
caught by neither: 1020 (56%) of 1829
F
De novo only, recall at top 1%
3
1%
23
5%
2
0%
caught by neither: 397 (93%) of 425
Which changes are flagged
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
G
All substitutions
antiGen (multi−tree) 0.84
antiGen (single−tree) 0.81
Where x which 0.73
Frequency (memorization) 0.71
Substitution matrix 0.68
PSSM (sequence composition) 0.66
Where only 0.40
Codon accessibility 0.29
Uniform 0.10
0.00 0.25 0.50 0.75
Recall at top 10%
H
De novo only
antiGen (multi−tree) 0.52
antiGen (single−tree) 0.39
Where x which 0.50
Frequency (memorization) 0.05
Substitution matrix 0.47
PSSM (sequence composition) 0.06
Where only 0.16
Codon accessibility 0.34
Uniform 0.10
0.0 0.1 0.2 0.3 0.4 0.5
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
I
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−1.0 −0.5 0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
J
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−0.5 0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
Temporal holdout (the main figure's regime)
NLL per mutation
K
All substitutions
antiGen (multi−tree) 2.18
antiGen (single−tree) 2.00
Where x which 1.63
Frequency (memorization) 1.31
Substitution matrix 1.29
PSSM (sequence composition) 0.79
Codon accessibility 0.61
Where only 0.32
Uniform 0.00
0.0 0.5 1.0 1.5 2.0
Nats better than uniform
L
De novo only
antiGen (multi−tree) 0.15
antiGen (single−tree) −0.52
Where x which 0.74
Frequency (memorization) −0.26
Substitution matrix 0.74
PSSM (sequence composition) −1.36
Codon accessibility 0.57
Where only 0.04
Uniform 0.00
−1.5 −1.0 −0.5 0.0 0.5
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
M
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−3 −2 −1 0
Multi−tree minus comparator, NLL per mutation
N
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−1 0 1
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Recall at top 10%
O
All substitutions
antiGen (multi−tree) 0.88
PSSM (sequence composition) 0.88
antiGen (single−tree) 0.85
Frequency (memorization) 0.81
Where x which 0.79
Substitution matrix 0.71
Where only 0.47
Codon accessibility 0.31
Uniform 0.11
0.00 0.25 0.50 0.75
Recall at top 10%
Q
De novo only
antiGen (multi−tree) 0.52
PSSM (sequence composition) 0.07
antiGen (single−tree) 0.37
Frequency (memorization) 0.06
Where x which 0.50
Substitution matrix 0.52
Where only 0.16
Codon accessibility 0.37
Uniform 0.10
0.0 0.1 0.2 0.3 0.4 0.5
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
S
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
0.0 0.5 1.0 1.5
Multi−tree minus comparator, recall at top 10%
U
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−1 0 1
Multi−tree minus comparator, recall at top 10%
NLL per mutation
P
antiGen (multi−tree) 2.34
antiGen (single−tree) 2.20
Where x which 1.74
Frequency (memorization) 1.56
PSSM (sequence composition) 1.53
Substitution matrix 1.42
Codon accessibility 0.76
Where only 0.38
Uniform 0.00
0.0 0.5 1.0 1.5 2.0 2.5
Nats better than uniform
R
antiGen (multi−tree) 0.12
antiGen (single−tree) −0.67
Where x which 0.91
Frequency (memorization) −0.22
PSSM (sequence composition) −0.85
Substitution matrix 0.99
Codon accessibility 0.93
Where only −0.00
Uniform 0.00
−1.0 −0.5 0.0 0.5 1.0
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
T
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
−4 −2 0
Multi−tree minus comparator, NLL per mutation
V
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
H3N2 HA
H1N1pdm HA
hMPV B G
B/Victoria HA
Pooled
0 4 8
Multi−tree minus comparator, NLL per mutation
Pooled−validation epoch (deployed)
Per−tree best epoch
W
Epoch rule
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model
All substitutions
single−tree model
De novo only
where x which
All substitutions
where x which
De novo only
−1.0 −0.5 0.0 0.5 1.0
0.00 0.05 0.10 0.15 0.20
−0.05 0.00 0.05 0.10 0.15 0.20
Pooled estimate
X
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model
All substitutions
single−tree model
De novo only
where x which
All substitutions
where x which
De novo only
−0.5 0.0 0.5
−0.05 0.00 0.05 0.10 0.15
0.0 0.1
Pooled estimate